

Minimum Wage and Reallocation

Preliminary and Incomplete

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Roadmap

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Introduction 

Motivation: Why Reallocation Matters

- The idea that minimum wage reshapes *which* firms operate is not new.
 - ▶ The first modern minimum wage (New Zealand, 1890s) aimed to help “worthwhile companies” against “sweatshops” that undercut them (Nordlund, 1997).
 - ▶ Minimum wage as a policy to **improve the quality of firms** in the market.
- A common concern today: minimum wage harms small, low-productivity firms.
- But what if these exits *improve* aggregate productivity?
 - ▶ Less productive firms exit $\Rightarrow$ workers reallocate to productive survivors.
- Despite its long history, **causal evidence on reallocation has been scarce**.
 - ▶ Only two studies to date: Dustmann et al. (2022) on Germany; Rao and Risch (2025) on the U.S.
 - ▶ We provide evidence from South Korea.

- **Policy & Context**

- ▶ Introduction of a nationwide minimum wage in Germany (2015)
- ▶ Large and binding shock: $\sim 15\%$ of workers affected

- **Key Findings**

- ▶ Overall, no negative employment effects
- ▶ Low-wage workers reallocate to larger, higher-paying, more productive firms
- ▶ Regional avg. firm quality increases in more exposed areas

Rao and Risch (2025, QJE)

- **Policy & Context**

- ▶ Minimum wage increases in six U.S. states (2014)
- ▶ Independent businesses in highly exposed industries

- **Key Findings**

- ▶ Minimal employment effects; negative impact only on teenagers
- ▶ Firms finance higher wages through higher revenue and changing cost structures
- ▶ Positive selection through firm entry and exit increases industry productivity
- ▶ Workers reallocate from independent firms to large firms

Research Questions

- **RQ1.** What are the **employment** effects of the minimum wage shock?
 - ▶ Do we observe job losses, or adjustments along specific margins?
- **RQ2.** Does the minimum wage trigger worker **reallocation** across firms?
 - ▶ Do workers move to more productive, larger, or higher-wage firms?
 - ▶ Which workers experience stronger employment and reallocation effects?
- **RQ3.** What **mechanisms** generate this reallocation?
 - ▶ Firm exit (“cleansing”) and worker transition into larger, more productive firms.

Institutional Background and Data

Minimum Wage Trends in Korea

- National minimum wage revised on an annual basis.
- Period features sharp minimum wage hikes in 2018–2019, followed by a slowdown.

Year	Minimum Wage (KRW)	1-Year Growth Rate
2015	5,580	7.1%
2016	6,030	8.1%
2017	6,470	7.3%
2018	7,530	16.4%
2019	8,350	10.9%
2020	8,590	2.9%

- The large hikes in 2018–2019 generate meaningful variation in exposure.

Data: Employer–Employee Matched Panel

- **Administrative employer–employee matched panel (2015–2019).**
 - ▶ Tracks workers' employment histories, separations, and destination firms
 - ▶ Linked firm information includes average wages, firm size, and revenue per worker (productivity proxy).
- **Data issues:**
 - ▶ Working hours are not observed and are imputed using the "Survey on Labor Conditions by Employment Type", based on sex, age group, industry, and income.
 - ▶ Firms are identified at the firm (not establishment) level. Miss switchers between establishments within firms.
 - ▶ No information on self-employment. Miss (1) the reallocation channel from wage employment to self-employment (treated as *not working*) or (2) transitions from self-employed with employees to self-employed without employees (treated as *firm exit*).

Sample

- Private-sector firms: Excluding nonprofits and foreign entities
- Low-wage workers: Wages between 70% and 150% of the minimum wage
- Full-time workers: Excluding daily workers (measurement issues in working hours)
- Outliers removed based on wage growth and changes in months worked
- About 2.5 million workers per year observed
- MW bite = workers below the next year's minimum wage (about 13%)

Descriptive Statistics by MW bite

	Bite = 0	Bite = 1	Diff	S.E.
<i>Worker Characteristics</i>				
Age	42.17	44.96	2.791***	(0.011)
Female	0.604	0.481	-0.124***	(0.0004)
<i>Labor Market Status</i>				
Real monthly wage	1,539,828	951,780	-588,048***	(0.328)
Real hourly wage	9,725	6,459	-3,266***	(1.548)
Still employed	0.855	0.750	-0.105***	(0.0003)
<i>Firm Transition Outcomes</i>				
$\Delta \ln$ wage	0.060	0.228	0.169***	(0.0001)
$\Delta \ln$ firm productivity	0.027	0.019	-0.0008***	(0.0006)
$\Delta \ln$ firm size	0.004	0.046	0.043***	(0.0013)
$\Delta \ln$ firm wage	0.013	0.023	0.010***	(0.0002)
$\Delta \ln$ firm tenure	-0.031	-0.029	0.002***	(0.0005)
Observations	13,184,344	1,612,568	14,796,912	

Regression Model

- We estimate the following linear model:

$$y_{it} = \beta \text{bite}_{i,t-1} + X_{i,t-1}\Gamma + \alpha_i + \tau_t + \varepsilon_{it},$$

where $\text{bite}_{i,t-1} = \mathbf{1}\{w_{i,t-1} < MW_t\}$ and $X_{i,t-1}$ includes worker and firm characteristics such as firm size, firm type, industry, and worker tenure.

- **Employment Outcomes:**

- ▶ Working regardless of employer change ($\text{still_employed}_{it}$)
- ▶ Continuous measure of labor supply ($\text{Diff_worked_months}_{it}$): between $t-1$ and t

- **Reallocation Outcomes:**

$$\Delta \text{FirmY}_{it} = \frac{\text{FirmY}_{B,t-1} - \text{FirmY}_{A,t-1}}{\text{FirmY}_{A,t-1}}$$

- ▶ FirmY = revenue per worker (productivity proxy), firm size, average tenure, and average wage, measured at $t - 1$, not t

Main Results: Individual Reallocation 

RQ1: Employment Effect of MW

- Sample period: 2015–2019

	Still Employed	Diff. Months Worked	Unemployed	Unemployed t & t+1
MW bite	-0.0615*** (0.0005)	-0.278*** (0.0073)	0.0565*** (0.0006)	0.0404*** (0.0005)
Worker FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Clustered SE	Worker	Worker	Worker	Worker
Sample	2015–2019	2015–2019	2015–2018	2015–2018
Observations	14,796,912	14,796,912	11,358,078	11,358,078

RQ2: Reallocation Effect of MW

- Sample restricted to worker who are still employed (2 years consecutively)
- MW exposure increases wage growth and probability of industry changes.

	Wage Growth	Same Firm	Industry Change	Region Change
MW bite	0.195*** (0.0003)	-0.022*** (0.0007)	0.012*** (0.0005)	-0.0004 (0.0003)
Worker FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Clustered SE	Worker	Worker	Worker	Worker
Sample	Still Employed	Still Employed	Still Employed	Still Employed
Observations	12,486,592	12,486,592	12,486,592	12,486,592

RQ2: Reallocation Effect of MW

- Sample restricted to workers who moved firms (switchers)
- Workers transition to the firms with higher productivity and higher wage.

	Δ Firm Productivity	Δ Firm Wage	Δ Firm Tenure	Δ Firm Size
MW bite	0.0427*** (0.0130)	0.0635*** (0.0038)	-0.0002 (0.0110)	0.0347 (0.0211)
Worker FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Clustered SE	Worker	Worker	Worker	Worker
Sample	Mover	Mover	Mover	Mover
Observations	2,765,388	2,765,388	2,765,388	2,765,388

Heterogeneity by Age Group

	Age 20s	Age 30–40s	Age 50–60s
Still employed	-0.108*** (0.0015)	-0.063*** (0.0007)	-0.035*** (0.0009)
Diff in months worked	-0.342*** (0.0194)	-0.313*** (0.0099)	-0.181*** (0.0123)
Wage Growth	0.211*** (0.0007)	0.192*** (0.0004)	0.191*** (0.0004)
Stayed at same firm	-0.066*** (0.0020)	-0.014*** (0.0009)	-0.013*** (0.0011)
$\Delta \log(\text{Firm Productivity})$	0.041 (0.0255)	0.031 (0.0209)	0.0618*** (0.0205)
$\Delta \log(\text{Firm Wage})$	0.059*** (0.0077)	0.074*** (0.0061)	0.053*** (0.0059)
$\Delta \log(\text{Firm Tenure})$	0.008 (0.0210)	0.012 (0.0182)	-0.031* (0.0176)
$\Delta \log(\text{Firm Size})$	-0.054 (0.0432)	0.087*** (0.0307)	0.0613* (0.0367)

Heterogeneity by Industry

	Manufacturing	Highly Exposed Industries
Still employed	-0.085*** (0.0015)	-0.0520*** (0.0009)
Diff in months worked	-0.208*** (0.0204)	-0.292*** (0.0122)
Wage Growth	0.188*** (0.0008)	0.202*** (0.0004)
Stayed at same firm	-0.044*** (0.0018)	-0.021*** (0.0011)
$\Delta \log(\text{Firm Productivity})$	0.034 (0.0630)	0.019 (0.0240)
$\Delta \log(\text{Firm Wage})$	0.048** (0.0191)	0.047*** (0.0069)
$\Delta \log(\text{Firm Tenure})$	0.027 (0.0591)	-0.003 (0.0215)
$\Delta \log(\text{Firm Size})$	0.080 (0.0854)	-0.0008 (0.0461)

Note: Highly exposed industries include retail, food accommodation, and business facility management services.

Who Adjusts Through Job Loss vs. Reallocation?

Minimum wage shocks trigger different adjustment margins across groups

- **Employment Type Heterogeneity**

- ▶ Daily workers: large employment losses, little firm upgrading
- ▶ Regular workers: smaller job loss but strong job-to-job reallocation
- ▶ $\Rightarrow$ Adjustment occurs through *job loss vs. job upgrading*

- **Age Heterogeneity**

- ▶ Young workers (20s): largest employment effects & strongest firm upgrading
- ▶ Older workers: weaker mobility and weaker reallocation

- **Industry Heterogeneity**

- ▶ Manufacturing: larger employment contraction, limited upgrading
- ▶ Highly MW-exposed services: strong movement to higher-paying and more productive firms

RQ3: Mechanism of Reallocation

- **Sample Definition**

- ▶ Sample period: 2015–2019
- ▶ Firms that hired more than one full-time worker in sample period

- **Company-level Exposure Variable**

- ▶ Percent increase in the wage bill needed to comply with next year's minimum wage.

$$\text{CompanyBite}_{j,t-1} = \frac{\sum_{i \in j} \max\{MW_t - w_{i,t-1}, 0\} h_{i,t-1} m_{i,t-1}}{\sum_{i \in j} w_{i,t-1} h_{i,t-1} m_{i,t-1}}$$

- **Baseline Specification**

- ▶ We estimate the following baseline specification:

$$\text{Exit}_{jt} = \beta \text{ CompanyBite}_{j,t-1} + X_{j,t-1}\Gamma + \alpha_j + \tau_t + \varepsilon_{jt},$$

- ▶ Controls: worker composition, industry, company type, productivity

RQ3: Mechanism of Reallocation

- Reallocation patterns by Sample Groups.

	Total	Worker <30	Low Productivity	High Exposure
Bite	0.112*** (0.0128)	0.0169*** (0.0023)	0.0783*** (0.0094)	0.0559*** (0.0175)
R^2	0.475	0.615	0.589	0.484
Observations	5,856,070	5,626,539	1,955,311	3,661,999

Note: Highly exposed industries include retail, food accommodation, and business facility management services.

- Heterogeneity among small firms (fewer than 30 workers).

	Low	Mid	High
Bite	0.0188*** (0.0037)	0.0120** (0.0056)	0.0086 (0.0054)
R^2	0.716	0.734	0.681
Observations	1,871,157	1,883,059	1,872,323

Conclusion 

Main Takeaways

- **MW increases lead to both employment losses and worker reallocation.**
 - ▶ Workers reallocate toward larger, higher-paying, and more productive firms.
 - ▶ The adjustment margin exhibits strong heterogeneity by industry, employment type, and age.
- **Firm dynamics are central to the reallocation process in the labor market.**
 - ▶ Less productive firms are more likely to contract or exit following MW hikes
 - ▶ This pattern is consistent with a *cleansing effect*.
- **Future research directions.**
 - ▶ Previous studies used data from expansionary periods. We aim to examine how reallocation effects vary over business cycle.
 - ▶ We plan to extend firm dynamics analysis beyond exit to include *entry* patterns, and measure changes in regional aggregate productivity to assess overall productivity effects.

Thank you!

Comments and feedback are very welcome.